

Extended Technical Appendix

Physical AI as an Integrated Technology Stack: Trends in Foundational and Implementation Technologies for Embodied Intelligence

Excerpt from the paper : Strategic Considerations of the Gate-All-Around (GAA) Transistor for AI Semiconductors in Software-Defined Vehicles (SDVs) and Robotic Manipulation Systems: latest revised on Dec. 04, 2025

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Section-1. The advanced automotive semiconductor sector : RISC-V enhances SDV scalability and SDVs impact societal institutions.

On March 6, 2025, Infineon Technologies AG of Germany announced in an official press release that it will launch a family of automotive microcontrollers adopting the RISC-V architecture within the next few years.

This announcement was a strategic move aimed at ensuring flexibility, scalability, and portability in response to the era of Software Defined Vehicles (SDV). Specifically, by adding RISC-V as the “fourth pillar” alongside the existing AURIX™ (TriCore-based) and TRAVEO™ (Arm-based) product lines, the company pledged to lay a clear foundation for the future of automotive semiconductors.

Fundamentally, logic semiconductors require not only circuit design but also an instruction set architecture (ISA) that defines the basic instructions executed by the processor. The British company Arm provides both the Arm core as a circuit design and the ISA, and it continues to be adopted across a wide range of fields. In contrast, RISC-V is an ISA distinct from Arm's.

The major points that distinguish RISC-V from other ISAs are:

1. It is open source (and therefore does not incur licensing fees);
2. It enables collaborative R&D through shared development environments;
3. It has high customizability and extensibility in terms of algorithms;
4. It is modular in terms of algorithmic functionality

With the emergence of RISC-V, some experts predict that the revenue base of Arm—long established as a solid presence among semiconductor design companies—may be shaken in the future.

In 2025, there were some reports that Arm would begin a phased increase of up to 300% in its licensing fees. This adjustment applies primarily to Compute Subsystem (CSS) IP packages based on the Armv9 architecture, reflecting enhanced performance characteristics. The pricing strategy is rooted in the “Picasso Project,” initiated in 2019, through which Arm aims to expand beyond smartphones into markets such as PCs, Data centers, Robots, and Software-Defined Vehicles (SDVs).

However, the long-term support infrastructure—including debuggers, compilers, and the mature suite of development tools, all of which are associated with Arm's IP—is expected to maintain a competitive advantage over RISC-V for the foreseeable future.

Meanwhile, Intel, once dominant in the semiconductor industry has recorded six consecutive quarters of losses. Furthermore, in April 2025, at the China Shanghai Motor Show, Intel announced its second-generation AI-enhanced Software-Defined Vehicle (SDV) SoC and disclosed the signing of

strategic partnerships with Chinese companies. Infineon is deploying V2X-compatible security modules and in-vehicle Ethernet products for the Chinese market, while strengthening technical collaboration with Chinese Tier 1 suppliers such as Desay SV and Neusoft Reach.

Furthermore, Qualcomm is jointly developing V2X communication modules with Chinese OEMs — including BYD, Geely, and SAIC — through its Snapdragon Automotive Platforms.

This indicates that future automotive technology trends must be analyzed through the lens of semiconductor and communication technologies. Analyzing the transition to connected cars and SDVs requires advanced expertise in semiconductors, processor instruction set architectures (ISAs), and high-speed communication protocols.

The UK-based PwC (PricewaterhouseCoopers) defines SDV (Software Defined Vehicle) as “a foundation that connects the inside and outside of mobility through software and continuously provides users with value and experience.” Furthermore, it explains that SDV signifies not only user experience and corporate revenue, but also the reconstruction of mobility’s institutional and social structure.

There is a view that SDVs not only improve convenience for vehicle users, but also contribute to fair used car transactions by making vehicle usage more transparent. However, it does not stop there. Recently, Japanese private financial institutions, which hold as much as 190 trillion yen (1.28 trillion U.S. dollars) in deposits, announced a digital coin (DCJPY). Just like digital currency, which can visualize the circulation and usage of money, the essential value of SDVs lies not only in benefits for vehicle users, but also in the significant impact they have on social institutions.

Automotive technology is undergoing a quiet yet rapid and resolute transition in its dominant actors.

Section-2. Evolutionary trends in enhanced SDV (2nd Gen. AI) and BEV battery technologies — and China’s strategic involvement

The U.S. government announced its \$8.9 billion investment in Intel on August 22, 2025.

On August 18, 2025, four days prior to the U.S. government announcement, SoftBank Group announced a \$2 billion investment in Intel.

Intel's strategy to regain competitiveness is defined by two innovative technologies.

The first is its Backside Power Delivery Network (BSPDN), branded as PowerVia.

This innovation targets four principal objectives:

- ① Increased integration density
- ② Reduced power consumption
- ③ Accelerated signal processing
- ④ Uniform thermal distribution

As transistor geometries continue to shrink, PowerVia addresses the escalating interference between surface-level power and signal lines by relocating power delivery to the chip’s backside. This architecture improves power delivery efficiency and signal integrity, enabling more compact, performant, and thermally balanced designs.

The second thrust is Intel's Chiplet Chassis concept, a modular architecture enabled by the UCIe standard, which allows heterogeneous chiplets to be interconnected within a single package. This approach achieves a synthesis of high-performance computing and energy-efficient control, while supporting flexible configurations tailored to diverse application domains.

Specifically,

- x86 (CISC)-based CPUs
- RISC-V or Arm (RISC)-based accelerators and control units

By integrating both into a unified SoC, Intel assigns compute-intensive tasks to CISC cores and low-power control functions to RISC cores. This task division, combined with the reuse of existing RISC-V/Arm IP, shortens the design, manufacturing, and validation cycle, thereby accelerating development throughput and reducing cost. Around 2026 to 2027, there is a high possibility that UCIe-based products with a mix of CISC and RISC will be fully launched in the market. (UCIe outperforms PCIe in speed, latency, and power.)

As semiconductor chips become increasingly miniaturized and integrated, the trend toward higher die counts per chiplet is gradually becoming apparent.

Currently, the number of semiconductor chips (dies) contained within a single chiplet varies depending on the semiconductor design philosophy, product application, and packaging technology, which optimize the configuration.

In Intel's case,

for the data center GPU "Ponte Vecchio,"

a configuration containing up to three dies per chiplet is adopted.

Specifically, multiple dies separated by function are integrated:

the Compute Tile manufactured on TSMC N5,

the Rambo Cache manufactured on Intel 7,

along with the HBM (High Bandwidth Memory), interface, Xe-Link, and others.

Intel adopted a heterogeneous integration GPU for data centers.

On the other hand, the client-oriented CPU "Meteor Lake"

features CPU, Graphics, I/O, and SoC tiles each constructed on a single die,

maintaining a simple design where one chiplet equals one die.

In the case of NVIDIA,

the current "Hopper (H100)" employs a monolithic design and has not yet adopted chiplet technology.

However, the next-generation GPU "Rubin" is planned to utilize TSMC's SoIC technology for vertical stacking. Through integration with HBM (High Bandwidth Memory), it is anticipated to feature a configuration where one or more dies are contained within a single chiplet.

Fundamentally, SoIC vertical stacking targets thermal efficiency, signal latency reduction, and bandwidth optimization. While the exact number of dies remains undisclosed at this time, the use of multiple dies seems inevitable.

In AMD's server-grade EPYC Genoa CPUs, each CCD (Core Complex Die) functions as a single chiplet, with a basic configuration of one chiplet per die.

As Intel seeks to differentiate itself through advanced packaging technologies, Japan's Rapidus has surfaced as a candidate foundry for NVIDIA's upcoming 2nm products. Rapidus's core technology, Gate-All-Around (GAA), enables ultra-high transistor density, low power consumption, and versatile applicability. Meanwhile, major global semiconductor manufacturers are moving toward mainstream designs that stack and integrate multiple dies onto a single chiplet after 2025.

This structure, similar to CTP technology for BEV batteries, carries the risk that localized failures—such as a single battery cell malfunction—can propagate to affect overall performance. A failure in a single die can cause the entire chiplet to cease functioning. When a GPU consists of multiple chiplets, HBM, I/O, and control logic, the impact could significantly affect the GPU's overall processing capability. (Note: The failure mechanisms are distinct: Electrochemical/thermal for battery cells versus logical/physical for dies.)

In April 2025, at Shanghai Motor Show, Intel unveiled its second-generation AI-enhanced Software-Defined Vehicle (SDV) SoC, built on this multi-process-node integration. Concurrently, it announced strategic partnerships with Chinese firms ModelBest and Black Sesame Technologies, reinforcing its commitment to the intelligent automotive ecosystem.

In essence, the direction of technological innovation in SDV semiconductors and BEV battery systems is shared—both face challenges such as high-density integration (high performance), thermal management, energy efficiency, and productivity.

The speed of transformation in these technological domains is not only rapid, but despite being entirely different fields, the objectives of such transformation are strikingly similar.

Intel's development of products based on the RISC-V instruction set carries significant meaning as a movement to expand the options for instruction sets. This is because it is expected that Arm, which has held dominance through ISA licensing, will have its position threatened. The reason is that Arm's source of profit—"license fees obtained from SoC manufacturers for using the Arm instruction set"—may decrease in the future.

Since expansion of RISC-V-based products is expected going forward, it is predicted that not only existing Chinese SoC manufacturers (StarFive, Sophon) but also other SoC manufacturers will proceed with "departure from Arm dependence".

Also, while NVIDIA's source of profit is data center GPUs, some researchers point out that the greatest contributor behind that profit is CUDA.

CUDA provides APIs for repurposing the GPU's original graphics processing capabilities for numerical computation and machine learning, and defines explicitly in the CUDA API the synchronization control that is the "key" to the GPU's greatest selling point—parallel processing.

Furthermore, NVIDIA not only publishes the API, but since external AI developers can build and share their own libraries on the CUDA environment (NVIDIA Developer Program), considering that

NVIDIA manages the API implementation, the reality is that it forms a GPU-exclusive API platform of NVIDIA. In essence, AI development premised on CUDA indirectly supports the expansion of NVIDIA's GPU sales and profits.

Ultimately, just as the Cell-to-Pack (CTP) architecture in battery pack design surpasses the Cell-to-Module (CTM)—exemplified by the Volkswagen ID series (MEB platform)—in terms of energy density per unit pack volume, the growing adoption of products based on the RISC-V instruction set threatens the profit base of incumbent fabless semiconductor firms such as Arm (design house). Furthermore, NVIDIA's CUDA API library platform continues to robustly safeguard its GPU sales and profitability against the competitive advances of other high-performance semiconductor manufacturers.

One commonly cited weakness of NVIDIA is its high power consumption. Based on this fact, some experts predict that NVIDIA is unlikely to aggressively pursue logic semiconductors for devices like smartphones, outside of its core data center business. However, in battery electric vehicles (BEVs), which are equipped with large-capacity batteries, power consumption is not as critical a constraint as it is for smartphones. (BEVs have more power headroom than smartphones, but thermal, standby, and peak constraints still matter) As battery technology continues to advance and semiconductor power consumption decreases, it is possible that NVIDIA is also setting its sights on AI-controlled Humanoid Robots and Robotic Manipulation Systems, following its focus on BEVs.

In other words, high-performance Software-Defined Vehicles (SDVs) could emerge as a new market for advanced AI semiconductors.

This potential is evident when considering the precedent set by TSMC, which secured Apple's smartphone business as a major client. (Other weaknesses that have been pointed out include performance per watt, software optimization, and the overall strength of its platform, including modem integration)

In that context, since the automotive industry places high importance on stable supply and long-term support, manufacturing risks and potential delays are major concerns.

Given that NVIDIA is a fabless company relying on TSMC for production, Semiconductor suppliers for SDVs will be selected based on the reliability of their SoC platforms, and those SoC vendors must in turn rely on foundry partners capable of consistently delivering logic semiconductors to support long-term production.

Moreover, since power efficiency has a direct impact on the driving range of BEVs, Rapidus's 2nm GAA architecture—designed to minimize energy loss—offers a distinct advantage as a next-generation semiconductor platform for SDVs.

Rapidus has developed Raads (Rapidus AI-Assisted Design Solutions) and RUMS (Rapid and Unified Manufacturing Service), but Raads is not an application developer-oriented API like CUDA.

Raads is a design support framework for SoC manufacturers, not an open API platform for algorithm developers. It does not include support for logic construction or algorithm description. Similarly, RUMS does not provide APIs or support for logic structuring. Both are fundamentally unrelated to

CUDA-style abstraction. Put simply, both Raads and RUMS are not positioned as open API platforms intended for broader users of logic semiconductors, such as algorithm developers. Rapidus remains focused on hardware, while global players increasingly shift toward software ecosystems like RISC-V and CUDA. Despite its emphasis on high-performance, it appears disconnected from broader user needs. End-users, such as automotive manufacturers, must be able to select and implement compute modules optimized for their specific applications.

Rapidus needs to provide developer-facing APIs (for SDV developers) that encapsulate the system-level advantages of its GAA architecture—such as optimized compute primitives, memory control, DVFS tuning, and real time scheduling—rather than exposing raw transistor-level characteristics. To differentiate GAA, its electrical advantages over FinFET—such as superior threshold control, reduced leakage, and stable low-voltage operation—must be materially improved, and those GAA properties must be leveraged across (1) the transistor/PDK layer used by SoC designers, (2) the microarchitecture/runtime layer in the compiler domain, and (3) the algorithm layer in terms of compute performance, latency, efficiency, and determinism.

While GAA's physical advantages (the following four items) are clear, its performance per watt depends on microarchitecture, design, and implementation.

- * Improved compute efficiency by leveraging GAA's low leakage and fast switching characteristics.
- * Enhanced memory control through stable low-voltage SRAM behavior and reduced leakage.
- * Optimized voltage operation by utilizing precise threshold control for efficient DVFS tuning.
- * Improved real time responsiveness by taking advantage of fast switching and low jitter for deterministic execution.

The logical structure of artificial intelligence (AI) is broadly divided into **learning** (CUDA optimizes power usage during **training**) and inference. And, for semiconductors, the training process imposes a greater computational load and consumes more power compared to the inference process.

CUDA is optimized for training tasks to maintain high computational performance while minimizing power consumption. The key libraries available in the CUDA ecosystem, excluding the CUDA Core API, include cuDNN, TensorRT, cuBLAS, cuSPARSE, cuFFT, NCCL, Thrust, and the CUDA Graphs API. The CUDA Core API primarily handles GPU control, memory management, and kernel execution.

In contrast, TensorRT is an inference optimization library that accelerates inference through layer fusion, and expands a suite of inference-specific APIs in conjunction with cuDNN. In the inference process, the speed of response—that is, real time capability—is crucial, as it involves utilizing available information to derive conclusions through prediction and reasoning.

CUDA, developed by NVIDIA, makes APIs for both training and inference available to AI developers, supporting efficient deployment across semiconductor platforms, while GAA offers performance advantages over FinFET in both power efficiency and computational speed. That is to say, GAA represents an evolutionary enhancement of the physical foundation of logic semiconductors.

CUDA provides a suite of APIs applicable to both AI training and inference processes; however, AI

developers are particularly leveraging these APIs to optimize power consumption and computational load during the training phase.

Raads should be API-enabled and made accessible not only to SoC designers but also to algorithm developers at automotive manufacturers who use those SoCs to build proprietary service applications (SDV). Even as semiconductor and SoC manufacturing technologies advance, the absence of meaningful connection to logic algorithm developers renders them unusable and, therefore, valueless. **This reality has been proven by CUDA.**

In other words, without a bridge between silicon and software, even the most advanced node risks becoming a commodity. For Rapidus, the user will likely be NVIDIA, but in order to make AI algorithm developers its users in the future, it needs to act strategically.

As AI semiconductors configured with GPU and HBM are becoming mainstream, a quiet race for semiconductor supremacy is intensifying in AI-driven Software-Defined Vehicles (SDVs), connected car data centers, and BEV battery management systems (BMS)—marked by aggressive offense and resilient defense.

The fierce race to crown the next technological champion is already steadily proceeding.

Section-3. Biofidelity in AI-controlled Robotic Manipulation Systems for Advanced Humanoid Robots

The technology stack therefore extends beyond semiconductor hardware and software frameworks to the physical and control layers through which AI interacts with the real world.

Learning that humans acquire by interacting with the environment through their limbs is fundamentally different from AI, which relies solely on sensor input. Physical movement involves elements that cannot be fully abstracted—such as time delays, force transmission (physical action and reaction), friction, and unpredictable disturbances. It also includes environmental factors like changes in light sources (reflections, backlighting, shadows) that affect visual sensor performance, including color perception; thermal conduction that impacts infrared sensors; ambient acoustic conditions and strong winds that influence ultrasonic sensors; and materials like glossy black vehicle bodies, bumper resins, and carbon-black tires that interfere with LiDAR. (Ref ; In order to detect dynamic and abnormal behavior—such as a preceding vehicle’s tire detaching—with high precision using LiDAR, the following challenges exist: ①The material of the tire (black rubber that absorbs light) may weaken the reflection of LiDAR’s laser. ②The movement of the tire is fast and its bouncing behavior is complex, so depending on LiDAR’s scan cycle and resolution, it may be processed as “noise” and missed.)

Through interaction with the environment via the limbs, including tactile sensation, humans gain meaning grounded in action, which cannot be obtained through visual or auditory information alone.

China’s Ministry of Industry and Information Technology (**MIIT**) announced in 2023 the “Guiding Opinions on the Innovative Development of Humanoid Robots”, explicitly emphasizing breakthroughs in core technologies corresponding to the “cerebrum, cerebellum, and limbs”.

These are presumed to correspond respectively to AI algorithms (cerebrum), motion control systems (cerebellum), and high-performance actuators (limbs). Furthermore, China plans to initiate the formulation of national standards based on this technical architecture by 2025.

However, setting aside whether the technical architecture description published by China's MIIT accurately represents biological functions. When human biological activity is carefully observed, it becomes evident that humans routinely perform actions that far surpass the capabilities of current AI-controlled Robotic Manipulation Systems.

The human cerebrum functions as the command center for voluntary movement, setting goals and planning trajectories within an abstract space, much like an AI. The spinal cord executes selected responses under physical and temporal constraints, following instructions from the cerebrum. In contrast to this information transmission pathway, the cerebellum receives proprioceptive input from the spinal cord via the spinocerebellar tract (SCT).

This input conveys a subset of proprioceptive information about the state of muscles and joints, including cutaneous sensation, to the cerebellum, including biological state variables that do not rise to conscious awareness, such as sudden changes in load, pressure, deformation, temperature, acceleration, vibration, torque fluctuation, and acoustic stimuli, among others.

Using this unconscious data, the cerebellum engages in predictive and adaptive control resembling Model Predictive Control (MPC), continuously forecasting future states and optimizing its neuromotor execution schema (or muscle control signals) in real time.

By means of this cerebellum and the spinocerebellar tract (SCT), humans unconsciously and reflexively adjust their own actions. Accordingly, through embodied interaction with their environment, humans have evolved not only explicit knowledge but also adaptive heuristics—rules of thumb or embodied wisdom—that sustain biological integrity by aligning strategic behavior with intrinsic physical constraints.

Ref : When considering the application of Model Predictive Control (MPC) to robots and autonomous vehicles, MATLAB's MPC Toolbox (developed by MathWorks) is a well-known tool. While this toolbox is well-suited for abstract design, algorithm verification, and prototype development in control architecture research, it presents several challenges for use in mass-produced control systems. These include commercial deployment costs related to licensing, issues of real time performance and latency, memory constraints in embedded environments, compatibility with AUTOSAR-compliant abstraction layers, and compliance with safety standards such as ISO 26262. As a result, it is inferred that, at present, no research projects exist that seriously apply Model Predictive Control (MPC) with mass production and commercialization as a foundational premise.

Currently, MATLAB code can be translated into CUDA C/C++ and further optimized for embedded GPU deployment using GPU Coder, enabling high-performance execution on NVIDIA platforms. Additionally, Simulink models support automatic CUDA code generation, facilitating integration of control architectures—including Model Predictive Control (MPC)—into real time autonomous driving systems. This workflow allows for algorithm-level design and simulation within

MATLAB/Simulink, followed by deployment of parallelized, hardware-optimized control logic suitable for embedded automotive-grade GPUs.

For further explanation, CUDA serves as a parallel computing API that accelerates numerical computations—such as matrix operations, optimization, and differential equation solving—which are core components of MPC control algorithms and their underlying logic structures. It does not intervene in the control algorithm or logic structure inherent to MPC.

Regardless of whether Model Predictive Control (MPC) is essential for fundamental research on robots and autonomous vehicles, it is inferred that, unlike factory equipment robots, the commercialization of industrial products that leverage/incorporate human biological activity (**unconscious proprioceptive information processed by the cerebellum**) is necessary for care robots and autonomous vehicles, which operate in closer proximity to humans.

At the present, the function of the cerebellum is not limited to its classical role in predicting movement and feedforward correction; current neuroscience has revealed that it is also deeply involved in **non-motor cognitive functions** such as the regulation of emotion and attention.

By incorporating the physiological functions of the cerebellum, AI systems may acquire the capacity for predicting the behavior of others and learning and understanding social rules. The cerebellum should not be positioned merely as a “computational device for control”; rather, I estimate that it is the **non-motor cognitive functions** of the cerebellum which will serve as the **key technology** enabling autonomous driving systems and Humanoid Robots to achieve social acceptability.

For AI to truly enter the domain of **fully autonomous driving**, mere sensor simulation alone is insufficient. A decision-making model with physical embodiment is required—one that incorporates vehicle dynamic characteristics such as steering responsiveness, braking delay, inertia, and tire grip. In other words, for AI to genuinely and safely “drive”, **it must reconcile abstract judgment with physically constrained action selection**. Because abstract and physical spaces contain numerous orthogonal and often incompatible logical domains, the extent to which abstract space can complement physical space remains limited.

Current AI research in the domains of autonomous vehicles and robotics is primarily focused on Generative AI, which centers on the design of abstract reasoning spaces. Looking ahead, it might be necessary to establish a new research paradigm—Generative AI × Embodied AI (called Fusion Intelligence or Fusion AI)—that connects this abstract space design with Embodied AI, which emphasizes decision logic under physical constraints.

At the core of this new research paradigm lies the principle that genuine correctness resides not in virtual or simulated environments, but in the real world. Accordingly, the verification of autonomous driving and robotics systems cannot be socially accepted as sustainable technologies without reference to their dynamic performance under real world conditions. The reason is distinctly simple: **any system that fails under real world dynamics lacks legitimacy**.

Prior to the explanation provided by China’s Ministry of Industry and Information Technology

(MIT), the conceptual description of the E/E architecture for Software Defined Vehicles (SDVs), published in 2018, includes the following passage:

Several major elemental modules that constitute the Software Defined Vehicle (SDV) are grouped into functional domains, forming three to four layers. The form of these layers embodies characteristics unique to each technical concept or design philosophy. A spinal cord vertically penetrates multiple layers, enclosing a central nervous system that is protected by robust security—like that of a vertebrate. At the nodal points between the central and peripheral nervous systems, not only is information transmitted, but a strategic advisory role to the cerebrum is also assigned. As a result, advanced intelligence will dwell there, and it will likely be referred to as the AI-Gateway. That layer may also be given another name: the platform.

The **spinal reflex**, as a biological response, refers to a physiological phenomenon in which a reaction occurs immediately to external stimuli, using the spinal cord as the central processing unit without passing through the brain. In the conceptual description of the E/E architecture mentioned above, the nodal points between the central and peripheral nervous systems are capable of fulfilling functions equivalent to those of spinal reflexes. The result of an immediate reaction caused by a spinal reflex is fed back to the cerebellum, where the biological information is evaluated and learned. Through this process, the movements following the reflex are adjusted and corrected, thereby achieving smoothness and precision in motion.

Furthermore, in the cerebrum, the central structure known as the **basal ganglia**, in coordination with the cerebral cortex and the cerebellum, plays a core role in Embodied AI—the determination of the next action to be taken—through learning and training involving reward prediction and motivation. This is closely related to what was described in Section 3—the system’s **decision-making regarding actions** such as steering and braking, based on each vehicle’s unique responsiveness in steering and braking or its dynamic performance, while taking into account the **time budget** required to avoid obstacles.

If one dares to add further perspective, in the control of robots, research on operating systems (OS) is believed to be progressing in a manner similar to that of Software Defined Vehicles (SDVs). As robots are equipped with increasingly diverse functions, it is expected that the conditions of their operating environments will expand accordingly, and that both the types and number of sensors will increase. It is expected that, going forward, the fusion of Robotic Manipulation Systems (RMS) and Robotic Mobility Platform (RMP) will lead to increased societal demand for Mobile Manipulators capable of executing complex functions across diverse environments.

In such cases, it is presumed that research and development on Electrical/Electronic (E/E) architectures is also necessary, as is being done by automotive OEMs in the development of SDVs. This is because, as long as AI-controlled Robotic Manipulation Systems rely on statistical modeling of perception, cognition, and behavior derived from massive datasets, if the underlying physical constraint structures (E/E) and execution platforms (OS) are not treated as design targets, such robots may be regarded as not meeting **biofidelity** requirements in robotic technology.

Essentially, the E/E architecture acts like the human body's nervous system in its entirety.

Etymology: The term **biofidelity** is derived from “bio” (life) and “fidelity” (faithfulness), and refers to the concept of how faithfully a model replicates the behavior of actual biological systems.

In the service robot market, especially in Japan, Humanoid Robots that emphasize full-body **biofidelity** are not necessarily essential; rather, I consider it desirable to prioritize the research, development, and commercialization of robotic manipulation systems suited to diverse service forms where labor shortages are anticipated.

Just as with autonomous driving, the introduction of AI-controlled robotic manipulation systems into the service robot sector as a means of addressing social challenges is desired.

Specifically, I foresee it to be important to develop a dedicated operating system (OS) as the software core in robotic manipulation systems (RMS) and to establish a robust Electrical/Electronic (E/E) architecture as its physical foundation.

In Robotic Manipulation Systems (RMS), the component equivalent to the central nervous system is the Electrical/Electronic (E/E) architecture. In the signal pathways of motor control, it is essential to closely examine how the cerebellum and spinal cord integrate afferent signals from somatic sensors distributed throughout the entire system and mediate both consciously accessible, **linguistically expressible motor actions** and **pre-reflective, sub-symbolic motor behaviors** that elude verbal articulation. Should a logical extrapolation be allowed, I conjecture that how to handle pre-reflective, sub-symbolic neural signals that elude linguistic expression will become a key point in AI-based control.

*Reference note on **motor behavior**:

pre-reflective: A concept rooted in the phenomenology of Jean-Paul Sartre, referring to the layer of experience that is immediate and direct, occurring prior to the subject's reflective awareness or objectification of the self.

sub-symbolic: Refers to the level of sensory and motor information processing that precedes symbolic representation through language or logical description. In neuroscience and cognitive science, such processing is often attributed to the lower layers of neural networks.

Whenever I encounter the term sub-symbolic (or symbolic representation), I am invariably reminded of traffic safety experts who argue that mere adherence to traffic rules is all that is required to guarantee the preservation of real world driving safety.

*Reference note on **social acceptability**:

Embodied intelligence leverages predictive control based on a wide range of unconscious sensory inputs, and is equipped with heuristics that maintain biological coherence under physical and spatial constraints. In the case of humanoid deployment, collaboration and shared judgment with humans are essential. Moreover, for autonomous driving, the ability to respond to exceptional situations and interpret tacit social understandings becomes critical. These forms of judgment and control rely on empirical information (**intuition-based information without calculation or analysis**)

that cannot be obtained in virtual environments. I believe this is precisely what is required for Humanoid Robots and autonomous vehicles to be socially accepted in a pluralistic society where diverse human beings live.

Humanoid Robots and autonomous vehicles are required to achieve **social acceptability** by being integrated into the social structure not merely as tools, but as legitimate and truly autonomous “AI-controlled agents” within human social institutions. I believe it is insufficient to view AI control merely as a process of learning and inference; we must not overlook the crucial role played by the **basal ganglia** in behavioral decision-making (action selection and timing compensation).

This is because, for an AI system to be considered capable of social behavior, it is not sufficient for it merely to possess advanced learning and reasoning capabilities; it is also essential that it select its own actions and assume responsibility for the consequences of its actions.

In other words, it is individual decision-making rooted in responsibility that forms the foundation of sociality. **Thus, abduction and verification, not only passive learning, forge resolute decision-making. Therefore, for an AI to verify its own abductive inferences, it would require a physical embodiment.**

Section-4. Strategic SoC Collaboration between Intel and NVIDIA

Intel's x86 architecture is a typical CISC type, designed to execute complex instructions with a small number of commands, and capable of performing multiple operations with a single instruction. However, recent Intel CPUs internally decompose these instructions into RISC-like μ OPs (micro-operations) for execution. This means that while the instruction set remains CISC, the actual execution resembles RISC through the use of μ OPs. The μ OPs (micro-operations) only partially reflect RISC design philosophy, and many proprietary specifications exist without standardization. Therefore, it cannot become a standardized open-source (free) architecture like RISC-V. (Ref.: While the RISC-V instruction set is standardized, the cores themselves are not fully standardized and are implemented according to each company's proprietary specifications)

RISC-V is an ISA based on RISC principles, with simple and consistent instructions. It supports 32/64/128-bit and has a rich set of subsets, offering high extensibility. As a result, processors from SiFive, Western Digital's SweRV Core, and Infineon's automotive microcontrollers all adopt RISC-V.

If Rapidus aims to develop an autonomous driving SoC, it will require a different architecture from the Intel-NVIDIA alliance's new SoC <I+N>. Autonomous driving requires real time decision-making, and for that, dedicated AI processors are more efficient than GPUs. In such a case, the AI processor is placed at the center of the SoC, with a RISC-V-based CPU handling control and general-purpose processing. When there are many external sensors, such as in cases of heavy image processing and parallel computation loads, GPUs become the only efficient solution, so they remain as supporting components. However, the presence of expensive GPUs is not essential. (That said, for sensor fusion involving high-resolution LiDAR, high-sensitivity (high-brightness) multi-cameras, SLAM, and multimodal processing, GPUs remain effective)

Currently, Rapidus is collaborating with Esperanto and Tenstorrent, both of which can provide RISC-V IP and CPU cores. Rapidus is also working with Cadence, developing a design flow and IP portfolio compatible with AI/HPC 2nm GAA BSPDN. Therefore, it is technically capable of designing and manufacturing AI processors using the GAA 2nm process.

- * Performance : <I+N> excels in parallel processing via GPU, focusing on versatility;
<R+V> excels in real time performance with AI processor + RISC-V.
- * Efficiency : <I+N> GPUs consume high power and generate heat;
<R+V> AI processors are highly efficient and power-saving.
- * Cost : <I+N> GPUs are expensive, x86 (commercial IP) is costly;
<R+V> uses RISC-V and custom SoC design with no licensing fees.
- * Scalability : <I+N> NVLink and x86 are hard to customize;
<R+V> RISC-V is open and allows custom design.

Ultimately, <I+N> SoC is powerful for general-purpose AI/HPC, but in autonomous driving—a field requiring real time performance and OEM-specific processing—<R+V> SoC is more rational.

However, the new SoC from Intel and NVIDIA, through their semiconductor collaboration, not only features NVIDIA's top-tier GPU performance in parallel processing, but also benefits from a mature development environment of **x86 + CUDA + NVLink**.

- * x86 : Instruction set managed by Intel. License restrictions apply. Not open.
- * CUDA : NVIDIA's proprietary GPU parallel computing API cannot be used with other companies' GPUs..
- * NVLink : High-speed interconnect technology designed and controlled by NVIDIA. Specifications are not publicly available and not fully open standard.

sensors constitutes an effective countermeasure against NVIDIA's proprietary approach. Although abstraction layers such as APIs and SDKs do not lose direct compatibility with cutting-edge manufacturing process nodes like 3nm or 2nm, existing abstraction layers must be extended and adjusted accordingly.

* **Rapidus**: designs and provides an API group that abstracts the physical characteristics of GAA semiconductors (responsible for physical-layer abstraction in semiconductors as IDM).

* **Renesas**: designs an SDK that allows flexible combination of sensor groups and effective utilization of the Rapidus API group (responsible for functional integration layer in SoCs).

Through this structure, automotive OEMs aiming to advance SDVs can:

- ① effectively utilize the accurate abstraction of cutting-edge semiconductor physical characteristics and the definition of control boundaries such as power efficiency.
- ② freely select sensors optimal for SDV users.

The essence of technology lies in the power to break through the wall of <proprietary>, and that power comes from <standardization and openness>.

Section-5. The Problem of Semantic drift and "Sociality Acceptability" in Autonomous Driving and Humanoid Robots

Semantic drift refers to the phenomenon in which the meaning of words gradually changes over time or across different contexts. This is a natural evolution in human language use, and it is also closely related to the reconstruction of meaning in artificial intelligence.

In addition, in order to truly attain social acceptability, autonomous driving systems and Humanoid Robots will likely require a mode of thinking that is most difficult for AI.

That is, AI must carefully observe its surroundings through sensors, read the atmosphere and mood of the environment, and flexibly adapt to the surrounding situation. To do so, a mode of thinking known as **abductive reasoning** (or simply **abduction**) is necessary, formulating a hypothesis based on observed information, and then revising that hypothesis through intervention, i.e., action, to verify its validity. This is a form of reasoning distinct from both **deduction** and **induction**.

Abductive reasoning represents the ultimate challenge for AI, and overcoming this peak requires embodiment and dynamic interaction with the external environment.

***deduction**: A mode of reasoning in which specific conclusions are derived from general principles or laws (major premises). Because deduction is grounded in formal logic, it is conventionally regarded as a rigorous and strictly logical form of thought.

***induction**: A mode of reasoning that abstracts general laws from multiple individual cases or observations. While induction does not yield absolute certainty, it constitutes an empirical logic that provides the foundation for scientific discovery.

***abduction**: Although abduction may be considered a type of logical reasoning, it is not positioned as formal logic. Rather, it is understood as a creative and hypothetical logic, oriented toward the generation of explanatory hypotheses.

Ref.: In the early seventeenth century, when the geocentric model remained orthodox, Galileo Galilei refined the Dutch-invented telescope and conducted systematic astronomical observations. These observations led him to endorse Copernicus's heliocentric theory, for which he was tried by the Inquisition and sentenced to lifelong house arrest. His work exemplifies the methodological cycle of, observation, hypothesis, verification, and proof, thereby revealing the epistemic foundations of human inquiry.

As with humans, the formation of intelligence in AI must proceed without privileging any single mode of reasoning—abduction, induction, or deduction—over the others.

Social acceptability for autonomous vehicles does not equate to the ability to drive safely in the same way as a typical human driver. Likewise, social acceptability for Humanoid Robots does not correspond to the ability to walk or perform manual tasks in the same way as humans.

No matter how advanced the technology becomes in simulating the 'real world,' the resulting 'virtual space' inevitably retains the contours of the designer's **delusional** domain.

For example, in earthquake prediction, it is said that while it may be possible to forecast “regions with a high probability of future earthquakes,” it remains extremely difficult to predict “when an earthquake will occur.” Conversely, in weather forecasting, the challenge is not in predicting “the time when rain is likely to fall,” but rather in forecasting “which region will experience rainfall.” In other words, in light of inherent unpredictability, one resorts to maximum likelihood estimation based on probabilistic reasoning. Even if autonomous vehicles and Humanoid Robots are released from restricted environments such as factory premises and undergo extensive training within virtual spaces $\hat{=}$ the system designer's **delusion**, they cannot attain social acceptability merely through such simulations in isolation from real human diversity. Such training that utilizes virtual spaces induces the **illusion** among engineers that virtual space equals the real world.

Engineers who study autonomous driving and humanoids using AI-generated virtual environments must clarify which domains, in the temporal or physical spatial dimensions, lack sufficient unpredictability. Invoking unpredictability as a justification for unforeseen phenomena after the fact is also unacceptable.

***delusion:** The cause lies within the believer

— an internal misrecognition, false belief, or delusional conviction.

***illusion:** The misrecognition is induced by external mechanisms or conditions

— an externally triggered illusion or perceptual distortion.

I would like to reiterate that what humans have historically done in order to attain social acceptability is to engage in **abductive reasoning**, formulating and revising hypotheses through biologically grounded, biosensor-driven interaction with dynamic environments, and sustaining their lives through this continuous adaptive process. In other words, I believe that the loop of observation $\rightarrow$ hypothesis $\rightarrow$ spontaneous action (verification) $\rightarrow$ revision $\rightarrow$ proof constitutes the very essence of biological intelligence. In the real world, where not everything is digitized or verbalized, I

consider embodiment to be essential for an AI system to verify, through its own actions, the validity of the hypotheses, abductive reasoning (logical inferences), that it has formulated itself.

Humans, through bodily experience, have adapted to their individual environments by modifying the plausibility of their own hypotheses and continuously updating them to those most appropriate to themselves. It has become the optimal solution obtained through one's own body.

Through this process, humans acquired what is referred to as sociality and thereby came to be accepted by society.

While appearing superficially rational, ideologies and theories that essentially pursue self-serving interests are feared to undermine the fundamental value of sociality by influencing the concept of “social acceptability” in autonomous driving and Humanoid Robots. As a result, its original meaning becomes distorted, losing its essential integrity. That is to say, by being caught up in semantic drift, there is a possibility that the meaning of social acceptability will be gradually shifted away from its original essence. Consequently, discourse influenced by self-centered ideology poses a danger of undermining the social compatibility of technology.

Assuming that the essence of technology’s sociality lies in human happiness, the significance of Humanoid Robots and autonomous driving lies not merely in research and development, but rather in securing broader social compatibility and acceptability in the real world. Moreover, these technologies must deliver tangible value and benefits to humans in diverse real-world environments.

In Lieu of a Conclusion of EX-Appendix

Physical AI is not constituted by AI software algorithms alone; it emerges from the integration of foundational semiconductor technologies, computational architectures, software abstractions, operating systems, control architectures, physical embodiment, and mechanisms for interaction with the real world. Beyond this technological integration, Social Acceptability becomes an essential condition for the deployment and integration of Physical AI into society, grounded in real-world verification of its behavior and performance.

I affirm that I have no affiliations—direct or indirect—with any automotive corporation, semiconductor enterprise, political entity, or religious organization.

With kindest regards,

Akira.

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